



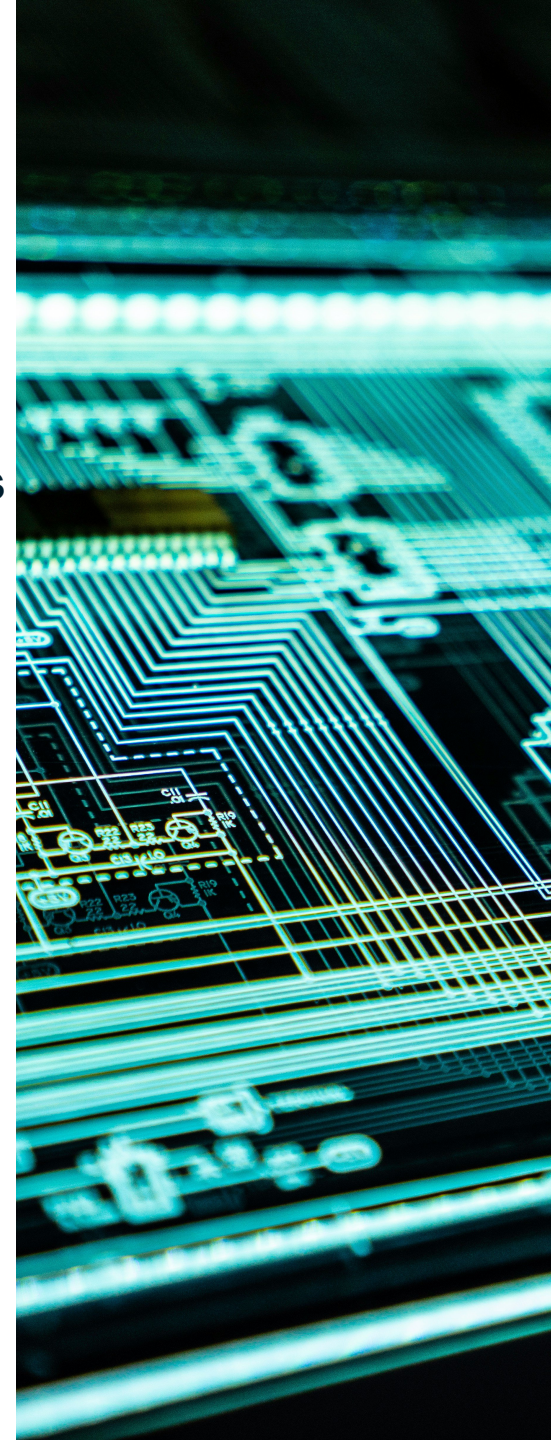
DIGITAL DOOR ACCESS SYSTEM USING COMBINATIONAL CIRCUITS

AGENDA

Traditional locks can be lost, damaged, or duplicated.

The aim of this project is to design a simple **digital password-based access**

- Verifies an entered password
- Allows access only for the correct password
- Denies access for an incorrect password
- Provides a clear visual indication of the access status



A DIGITAL DOOR ACCESS SYSTEM
IS AN ELECTRONIC SECURITY
SYSTEM THAT ALLOWS ACCESS
ONLY WHEN THE CORRECT
PASSWORD IS ENTERED.
IN THIS PROJECT, A **4-BIT
PASSWORD** IS ENTERED
THROUGH DIGITAL INPUTS AND
COMPARED WITH A PREDEFINED
PASSWORD USING LOGIC GATES.
CORRECT PASSWORD → ACCESS
GRANTED
INCORRECT PASSWORD →
ACCESS DENIED
THE SYSTEM IS SIMULATED USING
LOGISIM



OBJECTIVES OF THE PROJECT:

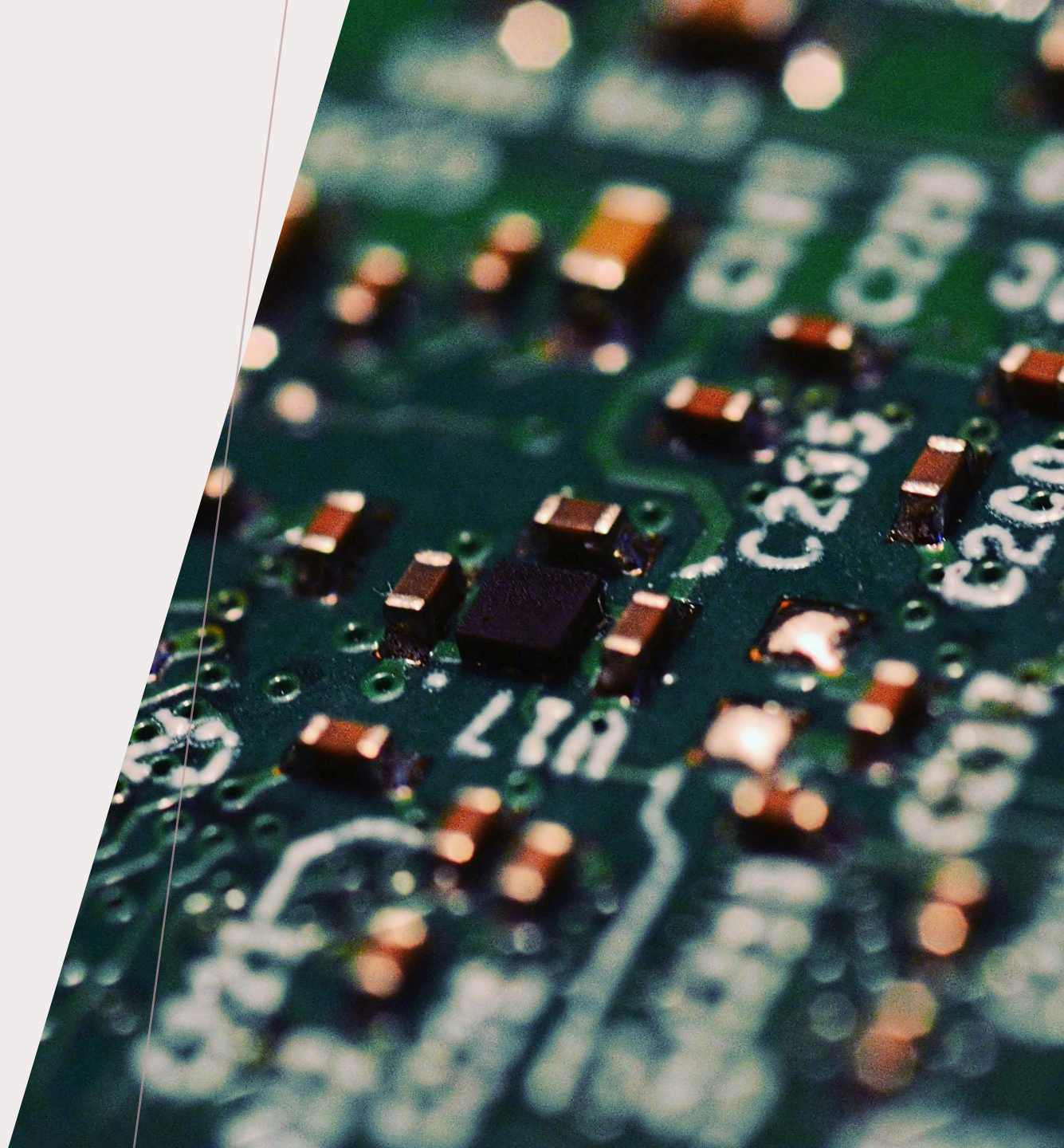
DESIGN A PASSWORD-BASED DIGITAL DOOR LOCK.

IMPLEMENT PASSWORD VERIFICATION USING LOGIC GATES.

SIMULATE THE CIRCUIT USING LOGISIM.

DISPLAY **ACCESS GRANTED** AND **ACCESS DENIED** CONDITIONS.

UNDERSTAND THE PRACTICAL APPLICATION OF COMBINATIONAL LOGIC.



Component	Purpose
Input Pins/Switches	Enter the 4-bit password
Constant	Store the predefined password
XNOR Gates	Compare each input bit
AND Gate	Check whether all bits match
NOT Gate	Generate access-denied signal
LEDs	Indicate access status
Output Pin	Represents door status
Logisim	Circuit design and simulation

COMPONENTS **USED**

The system uses **XNOR gates** to compare each bit of the entered password with the corresponding bit of the stored password.

For the project, the predefined password is:

1011

Each XNOR gate produces:

- **1** → Bits are equal
- **0** → Bits are different

The outputs of all four XNOR gates are connected to a **4-input AND gate**.

Therefore:

All bits match → AND output = 1 → Door opens

Any bit differs → AND output = 0 → Door remains locked

The access condition is:

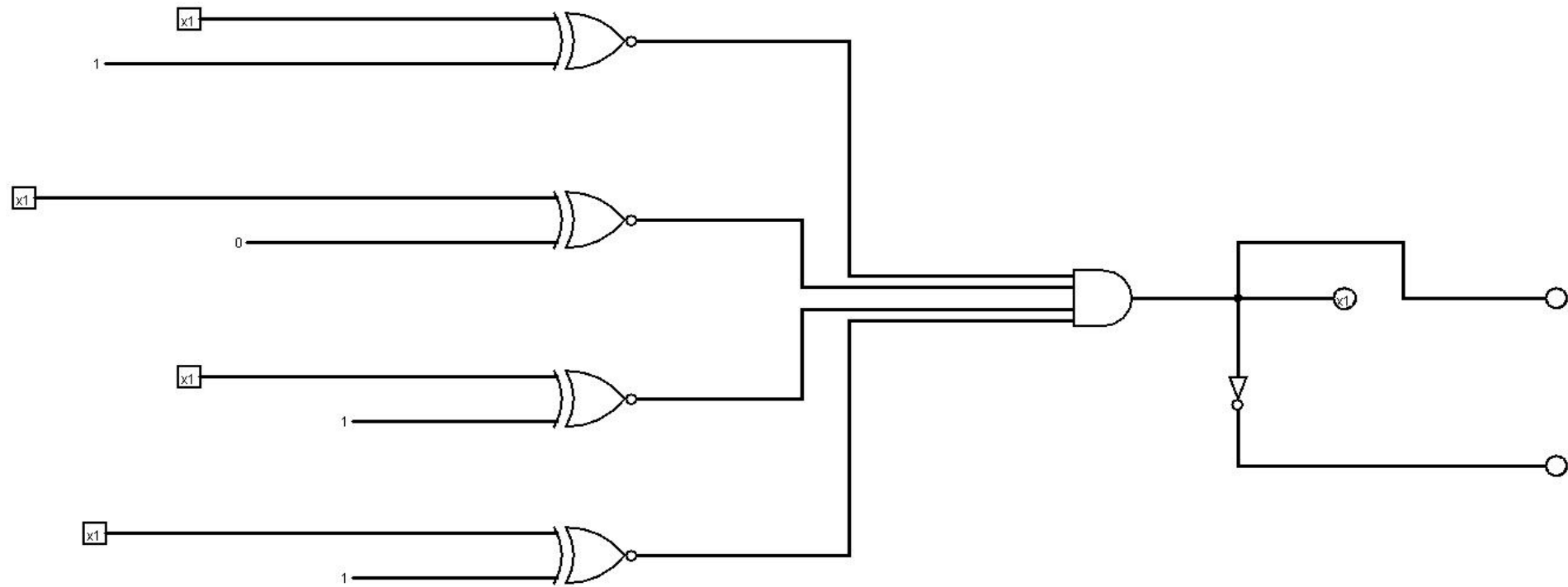
$\text{ACCESS} = (A \text{ XNOR } 1) \cdot (B \text{ XNOR } 0) \cdot (C \text{ XNOR } 1) \cdot (D \text{ XNOR } 1)$

LOGISIM SIMULATION:

Simulation Results

- The circuit was designed and simulated using Logisim.
- When the correct password **1011** is entered, the **Access Granted LED** turns ON.
- When an incorrect password is entered, the **Access Denied LED** turns ON.
- The door output remains LOW for incorrect passwords.

LOGISIM :



CONCLUSION

The **Digital Door Access System** demonstrates how digital logic can be used to create a simple security system.

A 4-bit password is compared with a predefined password using **XNOR and AND gates**. Access is granted only when all four bits match.

The project provides a basic foundation for developing more advanced electronic access-control systems.



THANK YOU